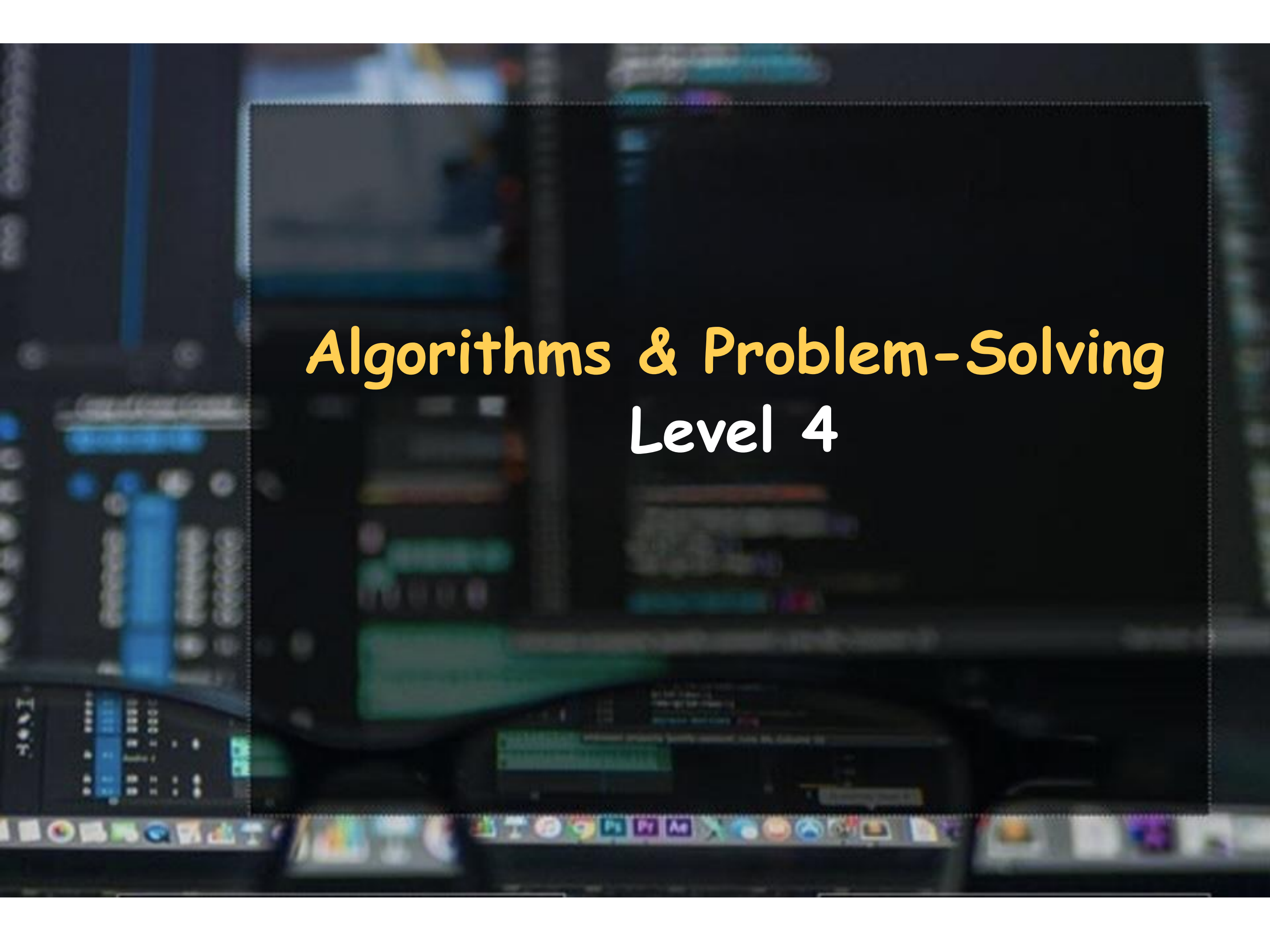
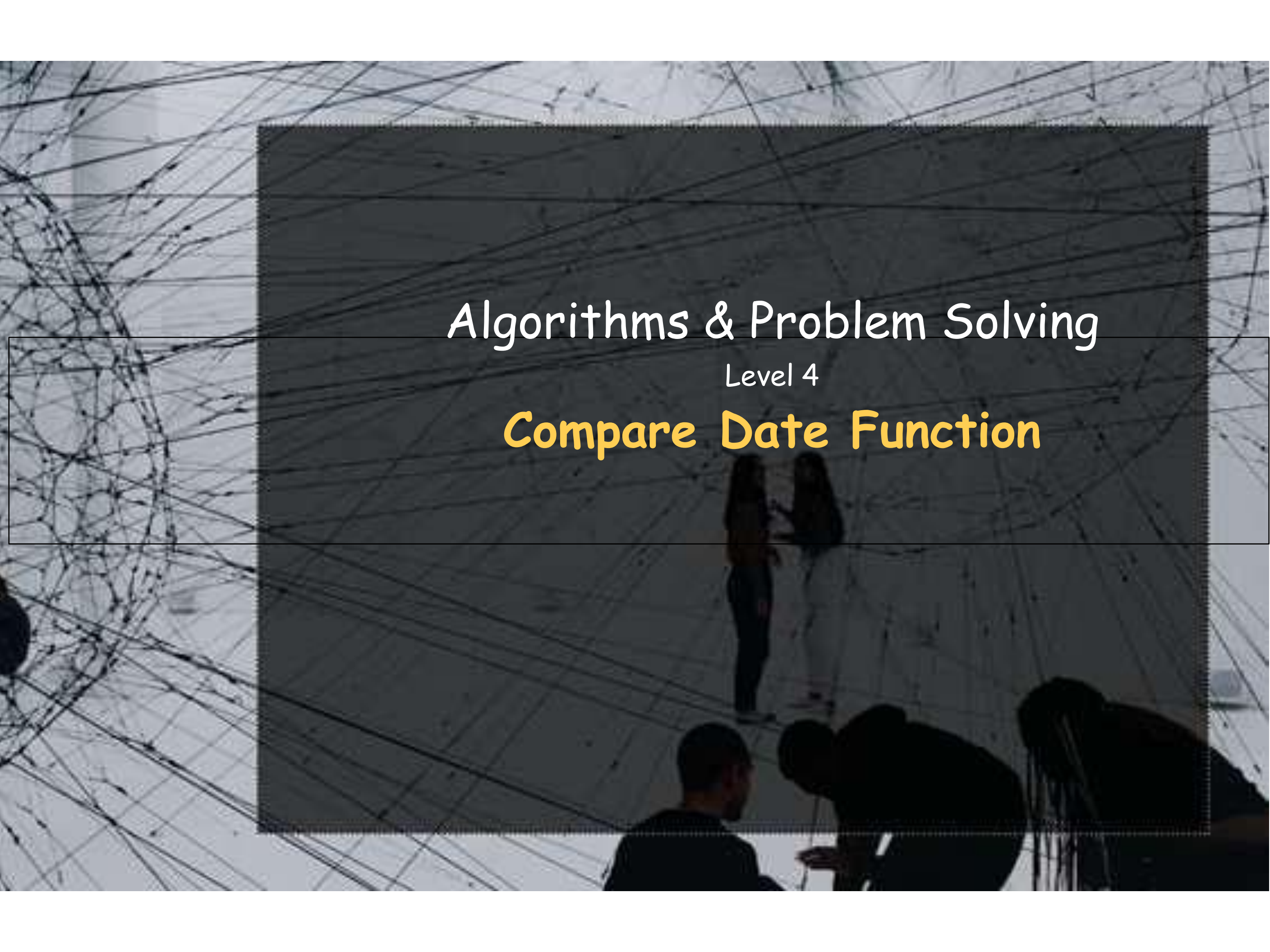




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a group, which adds a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Compare Date Function



Definition

- A program that reads two dates from the user and then compares them to determine if:
 - Date 1 comes before Date 2
 - Dates are equal
 - Date 1 comes after Date 2

Explanation:

- The comparison is based on chronological order (year, month, day).
- Logical functions (`IsDate1BeforeDate2`, `IsDate1EqualDate2`, `IsDate1AfterDate2`) are used to determine the relationship between the two dates.
- The basic function, `CompareDates`, returns an enum value representing the result.

Procedures

1 Defining the `struct sDate`

- Contains:

- Year

- Month

- Day

2 Defining the `enum enDateCompare`

- Definition: To determine the result of comparing two dates

- Before = -1

- Equal = 0

- After = 1

3 Helper functions for reading data

- ReadDay() - Inputs the day

- ReadMonth() - Inputs the month

- ReadYear() - Inputs the year

- ReadFullDate() - Combines the previous values into a single `sDate` structure

4 The `IsDate1BeforeDate2` function

- Definition: Checks if the first date precedes the second

- The comparison is performed first by year, then month, then day

5 IsDate1EqualDate2 Function

- Definition: Checks if two dates are exactly the same

6 IsDate1AfterDate2 Function

■ Definition: Checks if the first date is after the second, provided it is not before or equal to it.

7 CompareDates Function

■ Definition: Returns the value corresponding to the relationship between the two dates:

■ -1 The first date is before the second.

■ 0 The dates are equal.

■ 1 The first date is after the second.

8 Main() Function

■ Requires the user to enter two dates.

■ Calls CompareDates and prints the result.



Example:

Enter Date1:

Please enter a Day? 1

Please enter a Month? 1

Please enter a Year? 2000

Enter Date2:

Please enter a Day? 1

Please enter a Month? 1

Please enter a Year? 2022

Compare Result = -1

Summary

- ✓ The program compares two dates and returns an accurate time relationship.
- ✓ It uses a hierarchical order (year, month, day).
- ✓ It uses enum to make the code clearer and more organized.

Thank you 😊